

Surface Area Using Nets

Lesson 10-3

Name: _____

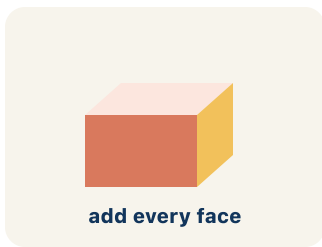
Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

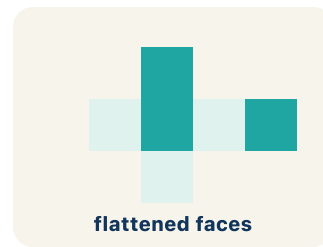


add every face

*A box $6 \times 4 \times 2$: $SA = 2(24) + 2(12) + 2(8) = 88 \text{ in}^2$
— like wrapping paper needed*

Surface area

The total area of all the flat sides of a solid.

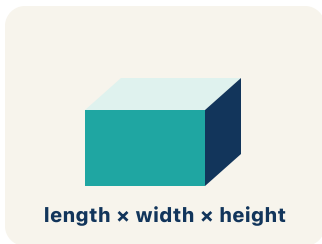


flattened faces

*Cut along the edges of a cereal box and unfold it
flat — you see all 6 faces*

Net

A flat shape that folds up into a solid.

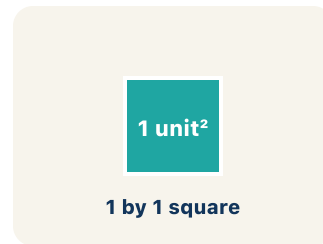


length $\times$ width $\times$ height

*A rectangular prism has 6 faces: top, bottom, front,
back, left, right*

Face

One flat side of a solid shape.

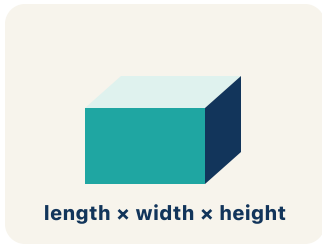


1 by 1 square

*A rectangle drawn on paper is 2D — it has length
and width but no depth*

Two-dimensional

A flat shape with length and width, but no thickness.



A cube has 12 edges — the lines where two flat faces connect

Edge

The line where two flat sides of a solid meet.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the surface area of a solid by using its net.

1 Notice

Your class is designing custom wrapping paper for time capsule boxes.

2 Key idea

I can find the surface area of a solid by using its net.

3 Words I'll use

Surface area

Net

Face

Two-dimensional

Edge



Think About It

- How many faces does a rectangular prism have?
- What shapes do you see when the box is unfolded?
- Are all the faces the same size?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A rectangular prism has $l = 6$ in, $w = 4$ in, $h = 2$ in. What is its surface area?

- A. 88 in^2
- B. 48 in^2
- C. 88 in^3
- D. 44 in^2

Step 1

$$SA = 2(6 \times 4) + 2(6 \times 2) + 2(4 \times 2) = 48 + 24 + 16 = 88 \text{ in}^2.$$



Answer: A. 88 in^2

 We do — together

Solve this one with your class using the same steps.

Problem: How many faces does a rectangular prism have?

- A. 6
- B. 4
- C. 8
- D. 12

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Surface area is measured in which type of unit?

- A. Square units (in^2 , cm^2 , ft^2)
- B. Cubic units (in^3 , cm^3 , ft^3)
- C. Linear units (in, cm, ft)
- D. No units needed

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

To find how much wrapping paper a time capsule box needs, why is unfolding it into a net more helpful than looking at the 3D box?

Level 1 support

Start here: A net lays every face flat so you can see and measure all six faces.

 **Need a hint?**

Sentence starters (optional)

A net helps because it shows all ____ faces laid ____.

Una red ayuda porque muestra las ____ caras puestas ____.

On the 3D box some faces are ____, but the net makes them all ____.

En la caja 3D algunas caras están ____, pero la red las hace todas ____.

Word bank: surface area net face two-dimensional edge

Listen for: Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

Level 2

Is there more than one correct net for the same box? Justify your thinking.

- There ____ be more than one net because ____.
- Different nets still give the same surface area because ____.

EXPLORE

For the box $l = 8$, $w = 5$, $h = 3$, you found three face pairs. How did you use the net to organize the faces into pairs, and which pair had the largest area?

Level 1 support

Start here: Opposite faces match in size, so the six faces form three equal pairs.

 **Need a hint?**

Sentence starters (optional)

The faces come in ____ pairs because opposite faces are ____.

Las caras vienen en ____ pares porque las caras opuestas son ____.

The largest pair is ____ $\times$ ____ = ____, counted twice for ____ in^2 .

El par más grande es ____ $\times$ ____ = ____, contado dos veces para ____ in^2 .

Word bank: **surface area** **net** **face** **two-dimensional** **edge**

Listen for: Students explain opposite faces are congruent (3 pairs), identify the top/bottom $8 \times 5 = 40$ as largest (80 in^2 for the pair), and total to 158 in^2 .

Level 2

How can you tell which face pair will be largest just by looking at the dimensions, before doing any multiplication?

- The largest pair uses the ____ two dimensions because ____.
- I can predict the pair sizes by comparing ____.

CONNECT

A gift box is $14 \times 10 \times 6$ in and one sheet of paper covers 400 in^2 . Talk through how to find how many sheets are needed to cover the whole box.

Level 1 support

Start here: Find the surface area first, then divide by what one sheet covers and round up.

 **Need a hint?**

Sentence starters (optional)

The surface area is ____ in^2 because $SA = 2(14 \times 10) + 2(14 \times 6) + 2(10 \times 6)$.

El área de superficie es ____ in^2 porque $SA = 2(14 \times 10) + 2(14 \times 6) + 2(10 \times 6)$.

I need ____ sheets because ____ $\div 400 \approx$ ____.

Necesito ____ hojas porque ____ $\div 400 \approx$ ____.

Word bank: surface area net face two-dimensional edge

Listen for: Students compute $SA = 280 + 168 + 120 = 688 \text{ in}^2$, then $688 \div 400 \approx 1.72$, and round up to 2 sheets to cover the whole box.

Level 2

Why must you round the number of sheets up instead of down, and what does the leftover paper represent?

- I must round up because ____.
- The leftover paper is about ____ in^2 because ____.

REFLECT

A classmate found the surface area of a box but only multiplied two of the three face pairs by 2. How would you explain why every pair must be doubled?

Level 1 support

Start here: Each of the three face shapes appears twice on the box, so all three areas must be doubled.

 Need a hint?**Sentence starters (optional)**

Every face pair must be multiplied by ____ because each shape appears ____ times.

Cada par de caras debe multiplicarse por ____ porque cada forma aparece ____ veces.

If you forget to double one pair, the surface area will be too ____.

Si olvidas duplicar un par, el área de superficie será demasiado ____.

Word bank: **surface area** **net** **face** **two-dimensional** **edge**

Listen for: Students explain a prism has opposite congruent faces (three pairs), so $SA = 2lw + 2lh + 2wh$, and missing a '2' undercounts one whole face.

Level 2

How does looking at the net make it obvious that there should be six faces, not five?

Generalize the rule.

- The net shows ____ faces because ____.
- In general, a rectangular prism always has ____ faces in ____ matching pairs.

Write About the Math

The Writing Revolution

I can explain my work using the words surface area, net, face, and two-dimensional.

1. Kernel Sentence subject + verb

Model: Surface area is the total area of all the flat sides of a solid.

Área de superficie es el área total de todos los lados planos de un sólido.

Write a kernel sentence about surface area. Use a subject and a verb.

Escribe una oración base sobre área de superficie. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Surface area matters in math

Área de superficie importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Surface area matters in math because ____.

Área de superficie importa en matemáticas porque ____.

but
pero

Surface area matters in math, but ____.

Área de superficie importa en matemáticas, pero ____.

so
entonces

Surface area matters in math, so ____.

Área de superficie importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about surface area.
Di un hecho verdadero sobre surface area.

Surface area ____.

Question
Pregunta

Ask a question about surface area.
Haz una pregunta sobre surface area.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about surface area.
Muestra entusiasmo sobre surface area.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with surface area.
Dile a un compañero qué hacer con surface area.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

A net helps because it shows all ____ faces laid ____.

Una red ayuda porque muestra las ____ caras puestas ____.

On the 3D box some faces are ____, but the net makes them all ____.

En la caja 3D algunas caras están ____, pero la red las hace todas ____.

The faces come in ____ pairs because opposite faces are ____.

Las caras vienen en ____ pares porque las caras opuestas son ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Why does a net help find surface area?

- A. It shows every face flat so you can add up their areas
- B. It hides some faces
- C. It changes the volume
- D. Nets only work for spheres

Show your work:

2. A rectangular box net shows faces of 6 cm^2 , 6 cm^2 , 8 cm^2 , 8 cm^2 , 12 cm^2 , 12 cm^2 . What is the surface area?

- A. 52 cm^2
- B. 26 cm^2
- C. 40 cm^2
- D. 36 cm^2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two boxes have the same volume of 60 in^3 . Box A is $5 \times 4 \times 3$. Box B is $10 \times 6 \times 1$. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

Sentence starter: Box A: $SA = 2(\text{---}) + 2(\text{---}) + 2(\text{---}) = \text{---} \text{ in}^2$. Box B: $SA = 2(\text{---}) + 2(\text{---}) + 2(\text{---}) = \text{---} \text{ in}^2$. Box --- uses less wrapping paper because --- .

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 9 \text{ in}$, $w = 5 \text{ in}$, $h = 3 \text{ in}$. What is the surface area?

- A. 174 in^2
- B. 135 in^2
- C. 174 in^3
- D. 87 in^2

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 6 — *A rectangular prism has 6 faces: top, bottom, front, back, left, and right.*
2. **You Try (notes):** A. Square units (in^2 , cm^2 , ft^2) — *Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.*
3. **Try It 1:** A. It shows every face flat so you can add up their areas — *A net unfolds the solid so all faces are visible to measure and add.*
4. **Try It 2:** A. 52 cm^2 — *Add all faces: $6+6+8+8+12+12 = 52 \text{ cm}^2$.*
5. **Exit Ticket:** A. 174 in^2 — *$SA = 2(9 \times 5) + 2(9 \times 3) + 2(5 \times 3) = 90 + 54 + 30 = 174 \text{ in}^2$. Surface area uses square units (in^2).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Surface area is the total area of all the flat sides of a solid.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).